SET [A]

Printed Page: 4 University Roll No:

Mid-Term Examination, Odd Semester: 2021-22

B.Tech (CEA), II Year, Semester III

BELH-0006: Ethics & Values

Time: 1: 30 hrs Max. Marks 30

---

## Section-A

Note: Attempt all questions/ Objective Type: .5X20= 10 Marks

1. Ethics is the study of:
   a. Law b. only Right c. only good and bad d. none of these
2. What do you mean by the valuable behavior?
   a. Utility b. Importance c. only profit d. ethical behavior
3. Critical thinking does not involve:
   a. Causes b. Reasons c. effects d. only Results
4. Problem-solving means:
   a. Only Thinking b. only Reasons c. Applying reasons in life d. none of these
5. Ethical attitude requires:
   a. Rationality b. Objectivity c. Values d. all of these
6. What is meta-ethics?
   a. Origin of ethics b. description c. Rules. d. Duties
7. Descriptive ethics does not mean:
   a. explaining morality b. rules c. principle of utility d. both b & c
8. Ethics that tell about the practices and norms:
   a. Normative b. Meta-ethics c. Descriptive d. All above
9. Only Skill development will produce?
   a. Good Engineer b. Bad Engineer c. Both Good & Bad Engineer d. none of these
10. Rational intelligence doe not presuppose:
    a. Only Technical skills b. emotional intelligence c. rationality d. reasons
11. The nature of Morality does not include:
    a. Personal reason b. objective reasons c. subjective reasons d. Individual reasons
12. Ethical Decision Making requires:.
    a. Five P model b. Rationality c. Problem solving d. All of these
13. Humility requires:
    a. Humbleness b. soft-skills c. rudeness d. none of these
14. Fundamental ethical values include:
    a. Humility b. Integrity c. Honesty d. All of these
15. Five P model is part of:
    a. Normative ethics c. meta-ethics c. descriptive ethics d. none of these
16. Ordinary decision making is based on :
    a. Desires b. Reason c. choice d. none of these

SET [A]

Printed Page: 4 University Roll No:

Mid-Term Examination, Odd Semester: 2021-22
B.Tech (CEA), II Year, Semester III
BELH-0006: Ethics & Values

Time: 1: 30 hrs Max. Marks 30

17. A good professional engineer requires:
a. Only Skills b. Only Values c. Both skills & values d. none of these
18. Values are extensive to:
a. Only personal life b. only professional life c. only social life d. all of these
19. Skills are having limited role in only:
a. Professional life b. personal life c. social life d. none of these
20. The violation of privacy is the case of :
a. Professional privacy b. Copy rights c. Right d. Informed Consent

## Section-B

Note: Attempt all questions 1X10=10 marks/ **Statement Based Questions:**
21. "Given the quantity of the two pleasures being same push-pin is as good as poetry". Which of the following statements are correct for the statement?

a. Mill said it b. quality of pleasure matters c. push-pin is higher pleasure d. None of these
22. Sumati Loves Ranjan that is why she was in trouble. She was working in the same company might have questioned her. She was upright professional. She should mind the ethics and profession. How should she resolve the problem?
a. By Break-up
b. By convincing Ranjan on ethical reasons
c. Minding only Profession
d. None of these
23. "It is better to be the dissatisfy Socrates than a satisfied human being". What can you infer from this statement?
a. Mill said it.
b. Quality difference
c. The person struggling for good is wrong
d. A& B
24. The study develops your attitude, your personal and professional life. You become rally a good person to understand the relationships. It is about the development of you morality also. It is about the:
a. The worth of ethics
b. The rules of ethics
c. The meaning of ethics
d. The study of meta-ethics.

SET [A]

Printed Page: 4 University Roll No:

Mid-Term Examination, Odd Semester: 2021-22
B.Tech (CEA), II Year, Semester III
BELH-0006: Ethics & Values

Time: 1: 30 hrs Max. Marks 30

25. "By the very human nature we desire pleasure and avoid pain".
a. It is psychological hedonism.
b. It is ethical hedonism .
c. It is both.
d. It is both not.
26. "Duties are different from desires as they are different from pleasures".
a. They are subjective.
b. They are obligatory.
c. They are objective.
d. Both objective and obligatory.
27. "Ethical decision making approaches are different in telling us what is right thing to do" because
a. Good result matters not all the time.
b. Sometimes duty that matters.
c. Sometimes result does not matters
d. all above
28. "Rachel will not be right if she ships without a proper check. However, in the situation if she does not her competitor will do". What is right thing to do for her?
a. Ship without check
b. Ship with check to ensure it does not harm human health.
c. Ship any how
d. Resign from the job.
29. "It's not utility or the duty but the good virtues can ensure the right decisions".
a. Virtues are based on pleasures
b. Virtues are based on duties.
c. Virtues represent maximum good.
d. A person do the right thing by good virtues.
30. You are in a dilemma whether to help your friend in cheating in the exam or just follow ethics and also save yourself being caught in doing unfair; what is right thing to do.
a. Follow only ethics don't mind friendship.
b. No help in exam
c. Help friend before the exam.
d. Be uncaring to save yourself.

## Section-C

Note: Attempt all questions : Multiple choice/ 1X10=10 marks
31. Was that person really good? What kind of ethical question is it?
1 Meta-ethical 2. Normative 3. Descriptive 4. All of these
a. Only 1 b. 1&2 c. 2&3 d. 3&4

SET [A]

Printed Page: 4 University Roll No:

Mid-Term Examination, Odd Semester: 2021-22

B.Tech (CEA), II Year, Semester III

BELH-0006: Ethics & Values

Time: 1: 30 hrs Max. Marks 30

---

32. Values are more extensive than skills because they work in
1 Only personal life 2. Only professional life 3. Only social life 4. None of these
a. only 1 b. only 2 c. only 3 d. only 4
33. The difference between animal and human is not based on
1 Pleasure & pain 2. desires and needs 3. rationality and choice 4. self interests
a. only 4 b. only 3 c. 1, 2 & 4d. only 3
34. Types of ethics are?
1. Normative & Selective 2. Descriptive & Normative 3. Meta-ethics 4. None of these
a. 2,3 b. only 1 c. only 3 d. only 4
35. Rational Intelligence means:
1. Applying only rationality 2. Applying only critical thinking 3. Applying problem-solving
4. All of these
a. only 2 b. only 1 c. only 3 d. 1,2
36. Ethical intelligence involves:
1. Rational Intelligence 2. Emotional Intelligence 3. Psychological intelligence 4. Skills
a. 1,2,3 b. only1 c. only4 d. only 1& 2
37. In car selling video professional ethics was shown by:
1. Man 2. Lady 3. Customer 4. none of these
a. only 4 b. only 1 c. 2& 3 d. only 2
38. What did the Mother teach in the trash collecting video?
1. Humility 2. Telepathy 3. Apathy 4. Utility
a. only 1& 2 b. only 2& 3 c. Only 3&4 d. only 1
39. Dog and bird fight in that video exemplifies:
1. Rationality & morality 2. Moral dillema3. moral duty 4. All of these
a. only 4 b. only 1 c. Only 1, 3 & 4 d. only 2& 3
40. The integrity video exemplifies:
1. Only honesty 2. Only Professional ethics 3. Only Integrity 4. All of these
a. only 4 b. only 1& 4 c. only 3 d. only 2 & 3

SET [B]

Printed Page: 4

University Roll No:

Mid-Term Examination, Odd Semester: 2021-22

B.Tech (CEA), II Year, Semester III

BELH-0006: Ethics & Values

Time: 1:30 hrs

Max. Marks 30

## Section-A

Note: Attempt all questions 1X10=10 marks/ Statement Based Questions:

1. "Given the quantity of the two pleasures being same push-pin is as good as poetry". Which of the following statements are correct for the statement?

a. Mill said it b. quality of pleasure matters c. push-pin is higher pleasure d. None of these

2. Sumati Loves Ranjan that is why she was in trouble. She was working in the same company might have questioned her. She was upright professional. She should mind the ethics and profession. How should she resolve the problem?

a. By Break-up
b. By convincing Ranjan on ethical reasons
c. Minding only Profession
d. None of these

3. "It is better to be the dissatisfy Socrates than a satisfied human being". What can you infer from this statement?

a. Mill said it.
b. Quality difference
c. The person struggling for good is wrong
d. A& B

4. The study develops your attitude, your personal and professional life. You become rally a good person to understand the relationships. It is about the development of you morality also. It is about the:

a. The worth of ethics
b. The rules of ethics
c. The meaning of ethics
d. The study of meta-ethics.

5."By the very human nature we desire pleasure and avoid pain".

a. It is psychological hedonism.
b. It is ethical hedonism .
c. It is both.
d. It is both not.

6. "Duties are different from desires as they are different from pleasures".

a. They are subjective.

SET [B]

Printed Page: 4

University Roll No:

Mid-Term Examination, Odd Semester: 2021-22

B.Tech (CEA), II Year, Semester III

BELH-0006: Ethics & Values

Time: 1:30 hrs

Max. Marks 30

b. They are obligatory.
c. They are objective.
d. Both objective and obligatory.
7. "Ethical decision making approaches are different in telling us what is right thing to do" because
a. Good result matters not all the time.
b. Sometimes duty that matters.
c. Sometimes result does not matters
d. all above
8. "Rachel will not be right if she ships without a proper check. However, in the situation if she does not her competitor will do". What is right thing to do for her?
a. Ship without check
b. Ship with check to ensure it does not harm human health.
c. Ship any how
d. Resign from the job.
9. "It's not utility or the duty but the good virtues can ensure the right decisions".
a. Virtues are based on pleasures
b. Virtues are based on duties.
c. Virtues represent maximum good.
d. A person do the right thing by good virtues.
10. You are in a dilemma whether to help your friend in cheating in the exam or just follow ethics and also save yourself being caught in doing unfair; what is right thing to do.
a. Follow only ethics don't mind friendship.
b. No help in exam
c. Help friend before the exam.
d. Be uncaring to save yourself.

Section B

Note: Attempt all questions/ Objective Type: .5X20= 10 Marks

11. Ethics is the study of:
a. Law b. only Right c. only good and bad d. none of these
12. What do you mean by the valuable behavior?
a. Utility b. Importance c. only profit d. ethical behavior
13. Critical thinking does not involve:
a. Causes b. Reasons c. effects d. only Results
14. Problem-solving means:
a. Only Thinking b. only Reasons c. Applying reasons in life d. none of these
15. Ethical attitude requires:
a. Rationality b. Objectivity c. Values d. all of these

SET [B]

Printed Page: 4

University Roll No:

Mid-Term Examination, Odd Semester: 2021-22

B.Tech (CEA), II Year, Semester III

BELH-0006: Ethics & Values

Time: 1:30 hrs

Max. Marks 30

---

16. What is meta-ethics?
    a. Origin of ethics b. description c. Rules. d. Duties

17. Descriptive ethics does not mean:
    a. explaining morality b. rules c. principle of utility d. both b & c
18. Ethics that tell about the practices and norms:
    a. Normative b. Meta-ethics c. Descriptive d. All above
19. Only Skill development will produce?
    a. Good Engineer b. Bad Engineer c. Both Good & Bad Engineer d. none of these
20. Rational intelligence doe not presuppose:
    a. Only Technical skills b. emotional intelligence c. rationality d. reasons
21. The nature of Morality does not include:
a. Personal reason b. objective reasons c. subjective reasons d. Individual reasons
22. Ethical Decision Making requires:
a. Five P model b. Rationality c. Problem solving d. All of these
23. Humility requires:
a. Humbleness b. soft-skills c. rudeness d. none of these
24. Fundamental ethical values include:
a. Humility b. Integrity c. Honesty d. All of these
25. Five P model is part of:
a. Normative ethics c. meta-ethics c. descriptive ethics d. none of these
26 Ordinary decision making is based on.:
a. Desires b. Reason c. choice d. none of these
27. A good professional engineer requires:
a. Only Skills b. Only Values c. Both skills & values d. none of these
28. Values are extensive to:
a. Only personal life b. only professional life c. only social life d. all of these
29. Skills are having limited role in only:
a. Professional life b. personal life c. social life d. none of these
30. The violation of privacy is the case of :
a. Professional privacy b. Copy rights c. Right d. Informed Consent

SET [B]

Printed Page: 4

University Roll No:

Mid-Term Examination, Odd Semester: 2021-22

B.Tech (CEA), II Year, Semester III

BELH-0006: Ethics & Values

Time: 1:30 hrs

Max. Marks 30

---

## Section-C

Note: Attempt all questions : Multiple choice/ 1X10=10 marks

31. Was that person really good? What kind of ethical question is it?
1 Meta-ethical 2. Normative 3. Descriptive 4. All of these
a. Only 1 b. 1&2 c. 2&3 d. 3&4

32. Values are more extensive than skills because they work in
1 Only personal life 2. Only professional life 3. Only social life 4. None of these
a. only 1 b. only 2 c. only 3 d. only 4

33. The difference between animal and human is not based on
1 Pleasure & pain 2. desires and needs 3. rationality and choice 4. self interests
a. only 4 b. only 3 c. 1, 2 & 4d. only 3

34. Types of ethics are?
1. Normative & Selective 2. Descriptive & Normative 3. Meta-ethics 4. None of these
a. 2,3 b. only 1 c. only 3 d. only 4

35. Rational Intelligence means:
1. Applying only rationality 2. Applying only critical thinking 3. Applying problem-solving 4. All of these
a. only 2 b. only 1 c. only 3 d. 1,2

36. Ethical intelligence involves:
1. Rational Intelligence 2. Emotional Intelligence 3. Psychological intelligence 4. Skills
a. 1,2,3 b. only1 c. only4 d. only 1& 2

37. In car selling video professional ethics was shown by:
1. Man 2. Lady 3. Customer 4. none of these
a. only 4 b. only 1 c. 2& 3 d. only 2

38. What did the Mother teach in the trash collecting video?
1. Humility 2. Telepathy 3. Apathy 4. Utility
a. only 1& 2 b. only 2& 3 c. Only 3&4 d. only 1

39. Dog and bird fight in that video exemplifies:
1. Rationality & morality 2. Moral dillema3. moral duty 4. All of these
a. only 4 b. only 1 c. Only 1, 3 & 4 d. only 2& 3

40. The integrity video exemplifies:
1. Only honesty 2. Only Professional ethics 3. Only Integrity 4. All of these
a. only 4 b. only 1& 4 c. only 3 d. only 2 & 3

SET [C]

Printed Page: 4 University Roll No:

Mid-Term Examination, Odd Semester: 2021-22

B.Tech (CEA), II Year, Semester III

BELH-0006: Ethics & Values

Time: 1:30 hrs Max. Marks 30

---

## Section-A

Note: Attempt all questions : Multiple choice/ 1X10=10 marks

1. Was that person really good? What kind of ethical question is it?
1 Meta-ethical 2. Normative 3. Descriptive 4. All of these
a. Only 1 b. 1&2 c. 2&3 d. 3&4
2. Values are more extensive than skills because they work in
1 Only personal life 2. Only professional life 3. Only social life 4. None of these
a. only 1 b. only 2 c. only 3 d. only 4
3. The difference between animal and human is not based on
1 Pleasure & pain 2. desires and needs 3. rationality and choice 4. self interests
a. only 4 b. only 3 c. 1, 2 & 4d. only 3
4. Types of ethics are?
1. Normative & Selective 2. Descriptive & Normative 3. Meta-ethics 4. None of these
a. 2,3 b. only 1 c. only 3 d. only 4
5. Rational Intelligence means:
1. Applying only rationality 2. Applying only critical thinking 3. Applying problem-solving 4. All of these
a. only 2 b. only 1 c. only 3 d. 1,2
6. Ethical intelligence involves:
1. Rational Intelligence 2. Emotional Intelligence 3. Psychological intelligence 4. Skills
a. 1,2,3 b. only1 c. only4 d. only 1& 2
7. In car selling video professional ethics was shown by:
1. Man 2. Lady 3. Customer 4. none of these
a. only 4 b. only 1 c. 2& 3 d. only 2
8. What did the Mother teach in the trash collecting video?
1. Humility 2. Telepathy 3. Apathy 4. Utility
a. only 1& 2 b. only 2& 3 c. Only 3&4 d. only 1
9. Dog and bird fight in that video exemplifies:
1. Rationality & morality 2. Moral dillema3. moral duty 4. All of these
a. only 4 b. only 1 c. Only 1, 3 & 4 d. only 2& 3
10. The integrity video exemplifies:
1. Only honesty 2. Only Professional ethics 3. Only Integrity 4. All of these
a. only 4 b. only 1& 4 c. only 3 d. only 2 & 3

SET [C]

Printed Page: 4

University Roll No:

Mid-Term Examination, Odd Semester: 2021-22
B.Tech (CEA), II Year, Semester III
BELH-0006: Ethics & Values

Time: 1:30 hrs

Max. Marks 30

---

Section B

Note: Attempt all questions 1X10=10 marks/ Statement Based Questions:

11. "Given the quantity of the two pleasures being same push-pin is as good as poetry". Which of the following statements are correct for the statement?

a. Mill said it b. quality of pleasure matters c. push-pin is higher pleasure d. None of these

12. Sumati Loves Ranjan that is why she was in trouble. She was working in the same company might have questioned her. She was upright professional. She should mind the ethics and profession. How should she resolve the problem?

a. By Break-up
b. By convincing Ranjan on ethical reasons
c. Minding only Profession
d. None of these

13. "It is better to be the dissatisfy Socrates than a satisfied human being". What can you infer from this statement?

a. Mill said it.
b. Quality difference
c. The person struggling for good is wrong
d. A& B

14. The study develops your attitude, your personal and professional life. You become rally a good person to understand the relationships. It is about the development of you morality also. It is about the:

a. The worth of ethics
b. The rules of ethics
c. The meaning of ethics
d. The study of meta-ethics.

15."By the very human nature we desire pleasure and avoid pain".

a. It is psychological hedonism.
b. It is ethical hedonism .
c. It is both.
d. It is both not.

16. "Duties are different from desires as they are different from pleasures".

a. They are subjective.
b. They are obligatory.
c. They are objective.
d. Both objective and obligatory.

17. "Ethical decision making approaches are different in telling us what is right thing to do" because

a. Good result matters not all the time.

SET [C]

Printed Page: 4

University Roll No:

Mid-Term Examination, Odd Semester: 2021-22

B.Tech (CEA), II Year, Semester III

BELH-0006: Ethics & Values

Time: 1:30 hrs

Max. Marks 30

---

b. Sometimes duty that matters.
c. Sometimes result does not matters
d. all above

18. "Rachel will not be right if she ships without a proper check. However, in the situation if she does not her competitor will do". What is right thing to do for her?
a. Ship without check
b. Ship with check to ensure it does not harm human health.
c. Ship any how
d. Resign from the job.

19. "It's not utility or the duty but the good virtues can ensure the right decisions".
a. Virtues are based on pleasures
b. Virtues are based on duties.
c. Virtues represent maximum good.
d. A person do the right thing by good virtues.

20. You are in a dilemma whether to help your friend in cheating in the exam or just follow ethics and also save yourself being caught in doing unfair; what is right thing to do.
a. Follow only ethics don't mind friendship.
b. No help in exam
c. Help friend before the exam.
d. Be uncaring to save yourself.

## Section-C

Note: Attempt all questions/ Objective Type: .5X20= 10 Marks

21. Ethics is the study of:
a. Law b. only Right c. only good and bad d. none of these

22. What do you mean by the valuable behavior?
a. Utility b. Importance c. only profit d. ethical behavior

23. Critical thinking does not involve:
a. Causes b. Reasons c. effects d. only Results

24. Problem-solving means:
a. Only Thinking b. only Reasons c. Applying reasons in life d. none of these

25. Ethical attitude requires:
a. Rationality b. Objectivity c. Values d. all of these

26. What is meta-ethics?
a. Origin of ethics b. description c. Rules. d. Duties

27. Descriptive ethics does not mean:
a. explaining morality b. rules c. principle of utility d. both b & c

SET [C]

Printed Page: 4 University Roll No:

Mid-Term Examination, Odd Semester: 2021-22

B.Tech (CEA), II Year, Semester III

BELH-0006: Ethics & Values

Time: 1:30 hrs Max. Marks 30

---

28. Ethics that tell about the practices and norms:
a. Normative b. Meta-ethics c. Descriptive d. All above

29. Only Skill development will produce?
a. Good Engineer b. Bad Engineer c. Both Good & Bad Engineer d. none of these

30. Rational intelligence doe not presuppose:
a. Only Technical skills b. emotional intelligence c. rationality d. reasons

31. The nature of Morality does not include:
a. Personal reason b. objective reasons c. subjective reasons d. Individual reasons

32. Ethical Decision Making requires:
a. Five P model b. Rationality c. Problem solving d. All of these

33. Humility requires:
a. Humbleness b. soft-skills c. rudeness d. none of these

34. Fundamental ethical values include:
a. Humility b. Integrity c. Honesty d. All of these

35. Five P model is part of:
a. Normative ethics c. meta-ethics c. descriptive ethics d. none of these

36 Ordinary decision making is based on :
a. Desires b. Reason c. choice d. none of these

37. A good professional engineer requires:
a. Only Skills b. Only Values c. Both skills & values d. none of these

38. Values are extensive to:
a. Only personal life b. only professional life c. only social life d. all of these

39. Skills are having limited role in only:
a. Professional life b. personal life c. social life d. none of these

40. The violation of privacy is the case of :
a. Professional privacy b. Copy rights c. Right d. Informed Consent

**Printed Pages: 2** **University Roll No.....**

**Mid-Term Examination, Odd Semester 2021-22**

**Programme:** B-Tech **Year:** II **Semester:** III

**Branch:** CS, AIML, CCV, DA, CSF and IIoT

**Computer Networks, BCSC008**

**Time : 2 Hours** **Maximum Marks: 30**

## Section- A

***Note: Attempt All Three Questions.*** 3 x 2 = 6 Marks

1. Name the four basic network topologies, and cite advantages of each type. For n devices in a network, what is the number of cable links required for each topology?
2. Which layers of the TCP/IP protocol suite are involved in a link-layer switch and router? How many times these layers process a frame/packet?
3. Assume a system uses five protocol layers. If the application program creates a message of 100 bytes and each layer (including the fifth and the first) adds a header of 10 bytes to the data unit, what is the efficiency (the ratio of application-layer bytes to the number of bytes transmitted) of the system?

## Section- B

***Note: Attempt All Three Questions.*** 3 x 3 = 9 Marks

1. Name three types of transmission impairment and their causes. What is the value of SNR and $SNR_{dB}$ for the noiseless channel?
2. CSMA is based on the principle "sense before transmit", Comment on it. Explain 1 persistent, non-persistent and p persistent method. A network has a data transmission bandwidth of $20 \times 10^6$ bits per second. It uses CSMA/CD in the MAC

layer. The maximum signal propagation time from one node to another node is 40 microseconds. What is the minimum size of a frame required to transmit over the network?

3. Explain the OSI reference model in detail. What are the three types of addresses used in Internet protocol stack (mention the size of each address)? Also discuss unicast, multicast and broadcast of a message.

## Section- C

***Note: Attempt Any Three Questions.*** 3 x 5 = 15 Marks

1. What is the total delay (latency) for a frame of size 5 million bits that is being sent on a link with 6 routers each having a queuing time of 5 μs and a processing time of 3 μs. The length of the link is 5000 Km. The speed of light inside the link is $2 \times 10^8$ m/s. The link has a bandwidth of 500 Mbps. Which component of the total delay is dominant? Which one is negligible?
2. Explain the working of CSMA/CA with the help of flowchart to avoid collision. Assume that there are four nodes (A, B, C and D) which are sending data in shared medium.
3. Compare and contrast a circuit-switched network and a packet-switched network.
4. How pure ALOHA differ from slotted ALOHA. Discuss the vulnerable time of Pure ALOHA and Slotted ALOHA. One hundred stations on a pure ALOHA network share a 1-Mbps channel. If frames are 1000 bits long, find the throughput if each station is sending 500 frames per second. How many frames will survive?

**Printed Pages: 3** **University Roll No.....**

**Mid-Term Examination, Odd Semester 2021-22**

**B. Tech. (EC, EE, EN, CE, CS), Year II, Semester III**

**Engineering Mathematics III (BMAS 1103)**

**Time: 2 Hours** **Maximum Marks: 30**

## Section- A

***Note: Attempt All Three Questions.*** 3 x 2 = 6 Marks

1. What are the normal equations to fit a straight line of the form

$$y = a + b x$$

to a given set of n values of x and y using method of Least squares?

2. If a random variable has a Poisson distribution such that

$$P(1) = P(2)$$

Find the mean of the distribution.

3. What is the first shifting property of Laplace transform? Use it to find

$$L[e^{-3t}\cos 4t]$$

## Section- B

***Note: Attempt All Three Questions.*** 3 x 3 = 9 Marks

1. Find Karl Pearson's coefficient of correlation between x and y:

| x | 1 | 3 | 5 | 7 | 8 | 10 |
|---|---|---|---|---|---|---|
| y | 8 | 12 | 15 | 17 | 18 | 20 |

2. (a) Find:

$$L\left\{\int_0^t e^t \frac{\sin t}{t} dt\right\}$$

(b) Using Laplace transform, evaluate:

$$\int_0^\infty e^{-3t}\, t \sin t\, dt$$

3. If 10% of the bolts produced by a machine are defective, Use binomial distribution to determine the probability that out of 10 bolts, chosen at random,

   (a) 1

   (b) None

   (C) At most two

   bolts will be defective.

## Section – C

***Note: Attempt Any Three Questions.*** 3 x 5 = 15 Marks

1. State the derivative property of Laplace transform. Using it, solve the differential equation:

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x = e^{-t} \sin t;\ x(0) = 0,\ x'(0) = 1$$

**OR,**

State convolution theorem. Hence or otherwise, evaluate:

$$L^{-1}\left\{\frac{p}{(p^2+1)(p^2+4)}\right\}$$

2. What is a periodic function? How can you find the Laplace transform of a periodic function $F(t)$ of period $T$ ? Find the Laplace transform of the following saw tooth wave function of period T:

$$F(t) = \frac{t}{T}, \qquad for\ 0 < t < T.$$

such that, $F(t+T) = F(t)$.

**OR,**

Find the inverse Laplace transform of $f(p) = log\left(1 + \frac{1}{p^2}\right)$

3. In a partially destroyed laboratory record of an analysis of a correlation data, the following results were legible:

Variance $of\ x = 9$
Regression equation: $10y = 8x + 66,\ 40x = 18y + 214.$
What were:

(a) The mean values of $x$ and $y$.

(b) The standard deviation of $y$.

(c) The correlation coefficient between $x$ and $y$.

4. By using $\chi^2$- test as a test of independence, find out whether there is any association between the income level and the type of schooling:

| Income | Public school | Government School |
|---|---|---|
| Low | 200 | 400 |
| High | 1000 | 400 |

[Given that: the tabulated value of $\chi^2$ at 5% level of significance for 1 degree of freedom is 3.841].

**Printed Pages:** **University Roll No.....**

**Mid-Term Examination, Odd Semester 2021-22**

**B.Tech (CSE/CCV/DA/IOT/AIML/CSF)**

**IInd Year, IIIrd Semester**

**BCSC1002: Object-Oriented Programming**

**Time: 2 Hours** **Maximum Marks: 30**

**Section- A**

**Note: Attempt All Three Questions.** **3 x 2 = 6 Marks**

Q1. Write short note on the following:
A) Encapsulation
B) Abstraction

Q2. What happens if we assign an integer value to a variable that is too large for that variable to hold?

Q3. Explain private, protected, public and default access modifiers in brief.

**Section- B**

**Note: Attempt All Three Questions.** **3 x 3= 9 Marks**

Q1. What is the difference between method hiding and method overriding explain with example?

Q2. What is the usage of final keyword with variable, method and class with suitable example?

Q3. Explain the difference between Conversion and casting with the help of an example?

**Section – C**

**Note: Attempt Any Three Questions.** **3 x 5 = 15 Marks**

Q1. What is inner class? What is the difference between inner class and subclass? Give an example of both an inner class and a subclass.

Q2. In a class hierarchy, in which order are the constructors for the classes that make up the hierarchy are called? How can a specific parameterised constructor of the superclass be called by the subclass?

Q3. What is an interface? Write a program to create an interface named "Show" which is used by class name "Student" having following structure:
In Show interface: -
Methods: - void showDetails ()
In Student class: -
Data Fields: - int age, String name, int reg_no;

Q4. All the banks operating in India are controlled by RBI. RBI has set a well defined guideline (e.g. minimum interest rate, minimum balance allowed, maximum withdrawal limit etc) which all banks must follow. For example, suppose RBI has set minimum interest rate applicable to a saving bank account to be 4% annually; however, banks are free to use 4% interest rate or to set any rates above it. Write a JAVA program to implement bank functionality in the above scenario and demonstrate the dynamic polymorphism concept. Note: Create few classes namely Customer, Account,
RBI (Base Class) and few derived classes (SBI, ICICI, PNB etc). Assume and implement required member variables and functions in each class.

**Printed Pages:** **University Roll No.....**

**Mid-Term Examination, Odd Semester 2021-22**

**B.Tech., 2nd Year, 3rd Semester**

**BCSC0003: Database Management Systems**

**Time: 2 Hours** **Maximum Marks: 30**

## Section- A

***Note: Attempt All Three Questions.*** 3 x 2 = 6 Marks

1. What is the difference between database state and database schema?
2. What are the responsibilities of Database Administrator?
3. What are the various problems that arise due to bad database design?

## Section- B

***Note: Attempt All Three Questions.*** 3 x 3= 9 Marks

1. Differentiate between stored vs derived attributes. What do you understand by recursive relationship? (3)
2. Describe the three-schema architecture. Why do we need mapping between schema levels? (3)
3. Let R= (A, B, C, D, E, F) be a relation schema with the following dependencies: C->F, E->A, EC->D, A->B. Find the key of R? (3)

## Section – C

***Note: Attempt Any Three Questions.*** 3 x 5 = 15 Marks

1. What are the set operation in relational algebra explain each and also explain what is the union Compatibility? (5)
2. What do you understand by data independence? What is difference between logical data independence and physical data independence? (5)

3. Consider the following schema, and write the following query in Relational Algebra (5)

Suppliers (Sid, sname, address)

Parts (pid, pname, color)

Catalog (sid, pid, cost)

I. Find the name of suppliers who supply some red part

II. Find the sid of supplier who supply red or green part both)

III. Find the sid of supplier who supply every part

IV. Find the avg cost

V. Delete record of that suppliers where sname is 'RAJ'

4. With a suitable example, describe 1st,2nd and 3rd Normal form. (5)

Printed Pages: 02 University Roll No.....

Mid-Term Examination, Even Semester 2021-22

B. Tech (CSE, All Specialization), II-Year, III Semester

Operating System (BCSC0004)

Time: 2 Hours Maximum Marks: 30

## Section- A

*Note: Attempt All Three Questions.* 3 x 2 = 6 Marks

1. Explain Non-preemptive and Preemptive CPU Scheduling with example. Elaborate all conditions when CPU scheduling is required.
2. Which CPU scheduling Algorithm causes Starvation? Write starvation and briefly explain its solution?
3. What is Process Control Block? Explain how PCB functions during context switch?

## Section- B

*Note: Attempt All Three Questions* 3 x 3 = 09 Marks

1. Distinguish Multiprogramming, Multitasking and Multiprocessing systems.
2. What is the role of kernel in OS? Distinguish clearly the difference between microkernel and monolithic kernel.
3. Explain Operating system as Resource manager! Clearly mention various components of Operating system.

## Section – C

*Note: Attempt Any Three Questions.* 3 x 5 = 15 Marks

1. During the state transition, in which states, the process remains in Secondary memory and main memory? Explain, Seven State process diagram with proper diagram.

2. Consider the following processes:

| Processes | Burst Time | Arrival Time | Priority |
|---|---|---|---|
| P0 | 7 | 0 | 3 |
| P1 | 3 | 2 | 2 |
| P2 | 1 | 3 | 0 |
| P3 | 4 | 5 | 1 |

Draw the Gantt Chart for SRTF (SJF-Preemptive) and Preemptive Priority CPU Scheduling. Also Calculate the average turnaround time, average waiting time and average response time. **(Priority:0>1>2>3)**

3. What is the role of schedulers? Which scheduler decide the degree of multiprogramming? Explain CPU scheduler with diagram.

4. Consider the following processes:

| Processes | Burst Time | Arrival Time |
|---|---|---|
| P0 | 7 | 0 |
| P1 | 4 | 2 |
| P2 | 3 | 5 |
| P3 | 6 | 10 |

Draw the Gantt Chart for Round Robin (Time Slice=2) and calculate the average turnaround time, average waiting time and average response time.

**Printed Pages: 2** **University Roll No.....**

**Mid-Term Examination, Odd Semester 2021-22**

**B.Tech. CSE/AIML/CSF/CCV/DA/IIoT, II Year, III Semester**

**BCSC1005: Computer Organization**

**Time : 2 Hours** **Maximum Marks: 30**

## Section- A

***Note: Attempt All Three Questions.*** 3x2=6 Marks

**Q1.** Convert:

a. $(6AB.CD)_{16} = (?)_8$ b. $(89.24)_{10} = (?)_5$

**Q2.** What is Von-Neumann architecture? Briefly explain the basic building blocks of the computer.

**Q3.** Differentiate

a) Multiplexer and De-multiplexer

b) Memory read and Memory write

## Section- B

***Note: Attempt All Three Questions.*** 3x3=9 Marks

**Q1.** Explain mathematically the design of carry look-ahead adder along with its advantage over the binary adder?

**Q2.** Represent the number $(-39.125)_{10}$ by using IEEE 32-bit and 64-bit floating point representation.

**Q3.** Perform the operation (-36)-(+23) by using 8-bit signed 2's complement method. Also, verify overflow exists or not.

## Section – C

***Note: Attempt Any Three Questions.*** 3x5=15 Marks

**Q1.** What is the need of stack organization? Explain Push and Pop operations. Convert the following arithmetic expression into reverse polish notation and show the stack operations for evaluating it.

7*5+(5+6)+(4*6+2)

**Q2.** What are the main fields of Instruction format? Write a sequence of instructions that will compute the following expression using two-address and one-address instruction.

F*G+(A*B+C)+D/E

**Q3.** Explain direct and indirect addressing mode. An instruction is stored at location 400 with its address field at location 401. The address field has the value 700. A processor register R1 contains the number 500. With reference to the addressing modes, identify the effective address in each of the following case.

a. Relative mode

b. Register indirect mode

c. Index mode with R1 register as index register

d. Immediate mode

**Q4.** Perform in steps the multiplication of the numbers (-15) and (-8) by using booth's algorithm.

**Printed Pages:03** **University Roll No.....**

**Mid-Term Examination, Odd Semester 2021-22**

**Course: B.Tech. (CSE), Year: II, Semester: IV**

# Software Engineering (BCSC 0009)

**Time : 2 Hours** **Maximum Marks: 30**

## Section- A

***Note: Attempt All Three Questions.*** 3 x 2 = 6 Marks

I. "Software Does not wear out but hardware does" explain the statement with the help of Bathtub curve.

II. Write any four major disadvantages of waterfall model.

III. What is the purpose of feasibility study? Explain any four parameters that plays an important role in conducting a feasibility study.

## Section- B

***Note: Attempt All Three Questions.*** 3 x 3= 9 Marks

I. The company wants to develop a software but requirements are not clear at initial phase and company is not so mature to adopt agile model. Which is the best development model in your opinion and why? Explain the model with the help of diagram.

II. What is requirement elicitation? Explain any four methods of requirement elicitation.

III. Create use case diagram for the following requirements:
Customer uses a bank ATM to check balances of his/her bank accounts, deposit funds, withdraw cash and/or transfer funds (use cases). ATM Technician provides maintenance and repairs to the ATM.

## Section – C

***Note: Attempt Any Three Questions.*** 3 x 5 = 15 Marks

I. A Restaurant management system Project is going to be developed in C language with the following parameters.

External Inputs: (a)14 with low complexity (b) 20 with average complexity (c) 23 with high complexity

External Outputs: (a)12 with low complexity (b)18 with high complexity

External Inquiries: (a) 13 with low complexity (b) 24 with average complexity (c) 25 high complexity

Internal logical files: (a) 27 with average complexity (b)17 with high complexity

External Interface files: (a) 29 with low complexity (b) 23 with high complexity

In addition to above, system requires

i.Average data communication ii. Performance is very critical iii. Designed code may be moderately reusable iv. System is not designed for multiple installation in different organizations. Other complexity adjustment factors are treated as Significant. Calculate the size of the project.

II. Consider the Restaurant management system Project (Same as given in previous Question No.1), The size of the project (You have already calculated in Question No.1). Use COCOMO to

1. Calculate Overall cost, schedule estimates, average staff size and productivity of the project (assume values for different Multipliers of different cost drivers are Nominal).
2. Calculate Cost & Schedule estimates for different phases.

III. Why Design of a software is important? How modularity plays an important role in designing a software? What is Module Coupling? Explain different types of Module Coupling with the help of an example.

IV. Consider am example of E-commerce website. Where customers can search the items, place the different orders, make payment. E-commerce company will search that item in its warehouse and supply the items, invoice to customers. Company also maintains the database of items ordered by customers, payment details, items available in warehouse. Create DFD of level-0 and level-1(show all the possible process, External Entity and Data Stores).

| Project Unit | Weighting factors | | |
|---|---|---|---|
| | Low | Average | High |
| External Inputs (EI) | 3 | 4 | 6 |
| External Output (EO) | 4 | 5 | 7 |
| External Inquiries (EQ) | 3 | 4 | 6 |
| External logical files (ILF) | 7 | 10 | 15 |
| External Interface files (EIF) | 5 | 7 | 10 |

| Project | $a_i$ | $b_i$ | $c_i$ | $d_i$ |
|---|---|---|---|---|
| Organic | 3.2 | 1.05 | 2.5 | 0.38 |
| Semidetached | 3.0 | 1.12 | 2.5 | 0.35 |
| Embedded | 2.8 | 1.20 | 2.5 | 0.32 |

Lifecycle Phase Values of $\mu_p$

| Mode & Code Size | Plan & Requirements | System Design | Detailed Design | Module Code & Test | Integration & Test |
|---|---|---|---|---|---|
| Organic Small S=2 | 0.06 | 0.16 | 0.26 | 0.42 | 0.16 |
| Organic medium S=32 | 0.05 | 0.16 | 0.24 | 0.38 | 0.22 |
| Semidetached medium S=32 | 0.07 | 0.17 | 0.25 | 0.33 | 0.25 |
| Semidetached large S=128 | 0.07 | 0.17 | 0.24 | 0.31 | 0.28 |
| Embedded large S=128 | 0.08 | 0.18 | 0.25 | 0.26 | 0.31 |
| Embedded extra large S=320 | 0.08 | 0.18 | 0.24 | 0.24 | 0.34 |

Lifecycle Phase Values of $\tau_p$

| Mode & Code Size | Plan & Requirements | System Design | Detailed Design | Module Code & Test | Integration & Test |
|---|---|---|---|---|---|
| Organic Small S=2 | 0.10 | 0.19 | 0.24 | 0.39 | 0.18 |
| Organic medium S=32 | 0.12 | 0.19 | 0.21 | 0.34 | 0.26 |
| Semidetached medium S=32 | 0.20 | 0.26 | 0.21 | 0.27 | 0.26 |
| Semidetached large S=128 | 0.22 | 0.27 | 0.19 | 0.25 | 0.29 |
| Embedded large S=128 | 0.36 | 0.36 | 0.18 | 0.18 | 0.28 |
| Embedded extra large S=320 | 0.40 | 0.38 | 0.16 | 0.16 | 0.30 |

**Printed Pages: 2** **University Roll No.....**

**Mid-Term Examination, Odd Semester 2021-22**

**B.Tech (AIML), Year : II, Semester : III**

**BCSE0701: Introduction to Machine Learning**

**Time: 2 Hours** **Maximum Marks: 30**

## Section- A

***Note: Attempt All Three Questions.*** 3 x 2 = 6 Marks

I. What are the differences between Classification and Regression?

II. What is a categorical data? Discuss one-hot encoding to convert categorical data to numerical data.

III. What do you mean by Overfitting? How to avoid this?

## Section- B

***Note: Attempt All Three Questions.*** 3 x 3= 9 Marks

I. Given the confusion matrix, find the Classification Accuracy, Recall, Precision, F-measure.

| | | Predicated | |
|---|---|---|---|
| | | Positive | Negative |
| Actual | Positive | 6 | 4 |
| | Negative | 2 | 8 |

II. What do you mean by the Logistic Regression? Explain the hypothesis representation and cost function for Logistic Regression.

III. Explain the concept of bias and variance tradeoff.

## Section – C

***Note: Attempt Any Three Questions.*** 3 x 5 = 15 Marks

I. What is the use of Regularization? Explain L1 and L2 Regularizations.

II. The values of independent variable **X** and dependent variable **Y** are given below:

| X | Y |
|---|---|
| 0 | 2 |
| 1 | 3 |
| 2 | 5 |
| 3 | 4 |
| 4 | 6 |

Find the least square regression line y = m*x + c. Estimate the value of **Y** when **X** is 10.

III. Explain the Gradient Descent Algorithm. How to choose the value of the parameter learning rate (α)?

IV. Consider a linear-regression model with N=3 and D=1 with input-output pairs as follows: y1=2, x1=1, y2=3, x2=1, y3=3, x3=2
What is the gradient of mean-square error (MSE) with respect to β1 (when β0=0 and β1=1)? Give your answer correct to two decimal digits.

Printed Pages: 2 University Roll No.....

**Mid-Term Examination, Odd Semester 2021-22**

**Programme:** B-Tech **Branch:** CCV **Year:** II **Semester:** III

**BCSE 1502: Introduction to Virtualization & Cloud Computing**

**Time : 2 Hours** **Maximum Marks: 30**

## Section- A

3 x 2 = 6 Marks

***Note: Attempt All Three Questions.***

1. What do you mean by hypervisor? Discuss various types of hypervisors.
2. Write down the advantages and disadvantages of storage area network (SAN).
3. Compare desktop virtualization with application virtualization?

## Section- B

3 x 3 = 9 Marks

***Note: Attempt All Three Questions.***

1. Classify virtualization based on the extent of hardware emulation? Also, discuss major layer of Xen server?
2. Differentiate between full virtualization and para-virtualization with examples.
3. Discuss different types of storage virtualization techniques.

## Section – C

3 x 5 = 15 Marks

***Note: Attempt Any Three Questions.***

1. What are the shortcomings of traditional IT infrastructure? Also, discuss how Virtualization and cloud helped to remove shortcomings of traditional IT infrastructure?
2. Identify the types of virtualization an organization must use to set up its offices in different part of the world. Justify your answer with proper reasoning. Also, discuss the cost impacts of proposed virtualization.
3. Summarize the working of VPN and VLAN with the help of example.
4. Explain the features of virtual desktop infrastructure. Why we need VDI?

**Printed Pages: 02** **University Roll No.....**

**First Term Examination, Odd Semester 2021-22**

**Program: B.Tech CSE (CSF) Year: II, Semester: III**

**BCSE-0601:IT Application & Data Security**

**Time : 2 Hour** **Maximum Marks: 30**

## Section – A

***Note: Attempt All Three Questions*** 3 x 2 = 6 Marks

1. What do you understand by threat and malware? Discuss and explain various application security principals.

2. Define Network eavesdropping. What are the parameters that determine the network eavesdropping vulnerability?

3. Differentiate between Authentication and Authorization with example. Discuss authentication vulnerability and its protection.

## Section – B

***Note: Attempt All Three Questions*** 3 x 3 = 9 Marks

1. What is information disclosure? Explain Information disclosure vulnerability and prevention methods.

2. What do you understand by input validation? Describe any two input validation vulnerability that can be exploited by an attacker.

3. How the same phases used in the traditional SDLC can be adapted to support the implementation of an information security project. Explain with example.

## Section – C

*Note: Attempt Any Three Questions.* 3x5=15

1. Alice and Bob wish to share private message, where each of them of generate two separate keys. How confidentiality, key management, authentication and integrity for the conversation between Alice and Bob will be achieved? Explain the strategy and also highlight the design issues related to the strategy proposed.

2. Using S-DES, show encryption of the given string 01000110 using the key 1010000110.

3. Managing a session during conversation is important. Discuss its importance and methods that can be deployed to ensure session management.

4. Write short note on following:
    a. Luring attack
    b. Cookie replay attack
    c. Brute force attack
    d. Elevation of Privilege

Printed Pages: 2 University Roll No.....

Mid-Term Examination, Odd Semester 2021-22

B.Tech (CSE(DA)), II$^{nd}$ Year, III$^{rd}$ Semester

BCSE0551: Introduction to Business Analytics

Time: 2 Hours Maximum Marks: 30

## Section- A

*Note: Attempt All Three Questions.* 3 x 2 = 6 Marks

1) List down at least four types of data measurement scales.
2) Define the link between different analytics capabilities.
3) Mention three different components of business analytics.

## Section- B

*Note: Attempt All Three Questions.* 3 x 3= 9 Marks

1) Draw a flow diagram for the Business analytics data-driven decision making.
2) Which of the following sequences have the highest Standard Deviation?
   2,4,6,8
   3,6,9,12
   4,8,12,16
   1,2,3,4
3) Find the median of a series of all the even terms from 4 to 296.

## Section – C

*Note: Attempt Any Three Questions.* 3 x 5 = 15 Marks

1) Given the following distribution of returns, determine the lower quartile:
{10% 23% 12% 21% 14% 17% 16% 11% 15% 19%}

2) A website displays 10 advertisements and the revenue generated by the website depends on the number of visitors to the site clicking on any of the advertisements displayed on the website. The data collected by the company has revealed that out of 2500 visitors, 30 visitors clicked on 1 advertisement, 15 clicked on 2 advertisements, and 5 clicked on 3 advertisements. Remaining did not click on any of the advertisements. Calculate:
   a) The probability that a visitor to the website will click on an advertisement.
   b) The probability that the visitor will click on at least two advertisements.
   c) The probability that a visitor will not click on any advertisements.

3) Suppose the fraction of undergraduate students who smoke is 15% and the fraction of graduate students who smoke is 23%. If one-fifth of the college students are graduate students and the rest are undergraduates, what is the probability that a student who smokes is a graduate student?

4) If the mean of numbers 28, x, 42, 78 and 104 is 62, what is the mean of 48, 62, 98, 124 and x?

Printed Pages: 13 University Roll No.....

**Mid-Term Examination, Even Semester 2021-22**

**B.Tech – CS (IIoT), II - Year , III - Semester**

**BCSE – 0902: Smart Industrial Connectivity**

**Time : 1 Hour** **Maximum Marks: 30**

***Note: Attempt All Questions.*** 60 x 0.5 =

30 Marks

**1.** Select the correct version for RPI

a) Raspberry pi 8
b) Raspberry pi 3B+
c) Raspberry pi 7
d) Raspberry pi -12B

**2.** Fill the Blanks pinMode(13,------);

a) Hello
b) OUTPUT
c) Print
d) None of above

**3.** Which of these are development boards

a) NodeMCU
b) Atmega328
c) STM32
d) None of above

4. Which of these can be used to measure distance

a) Ultrasonic sensor
b) LDR sensor
c) IR sensor
d) PIR sensor

5. Which of these are temperature sensor

a) LM35
b) LM345
c) LM3335
d) LM125

6. Which Sensor we used as Vibration detection sensor in Smart EV

a) Vibration Sensor
b) GY251
c) MPU6050
d) magneto Sensor

7. Which of these are true for Hardware protocols

a) SPI
b) TCP/IP
c) MQTT
d) APPL

8. A request from client is basically made of _____

a) Method
b) Task
c) Event
d) Signal

**9.** The Raspberry Pi is defined as the?

a) Mini computer
b) Micro Computer
c) Mega Computer
d) Nano Computer

**10.** Raspberry Pi consists of a _____ quad-core processor or microprocessor.

a) 16-bit
b) 32-bit
c) 64-bit
d) 128-bit

**11.** Which of the following is true for Python?

a) It is considered superior at software applications.
b) It was developed to encourage basic learning for computer science students and other growing countries.
c) It promotes Scratch and Python as the chief programming language
d) All of the above

**12.** How much RAM, the Raspberry Pi 3B+ has?

a) 2GiB of RAM
b) 1GiB of RAM
c) 4GiB of RAM
d) 8GiB of RAM

**13.** How many Input/output pins on board Raspberry Pi3 has?

a) 20
b) 30
c) 40
d) 50

14. Which operating system Raspberry Pi has?

a) Raspbian OS
b) OpenBSD
c) NetBSD
d) All of the above

15. Does Raspberry pi has ROM on Board

a) Maybe
b) Yes
c) NO
d) None of Above

16. What are the parameters that are default values?

a) Port_Name and Bits
b) Speed and Port_Names
c) Speed and Parity
d) Stop bit and Flow Control

17. LM35 can measure temperature in range

a) -55 to 150
b) -60 to 100
c) 0 to190
d) -90 to 100

18. RFID stands for

a) Radio frequency Identification
b) Ram Fiaona Identity
c) Random Field Generation
d) None

**19.** At What frequency Our RFID sensor Works

a) 2.45 GHZ & 5.4 GHZ
b) 860 – 960 MHZ
c) 433 MHZ
d) None of above

**20.** Which is True for GPS?

a) Global Positioning System
b) Global Planning System
c) Time and Ranging System
d) None

**21.** What type of Battery is used in Our Electric Vehicle?

a) Lithium ion
b) Lead acid
c) Lithium Polymer
d) None of above

**22.** What type of Motor is Used in our electric Vehicle?

a) Induction Motor
b) Synchronous Motor
c) Brushless DC Motor
d) DC geared Motor

**23.** What kind of signal is generated to control the speed of EV?

a) Digital signal
b) Analog signal
c) PWM signal
d) None of above

**24.** Which device is used to control the Speed of Electric vehicle

a) Electric Speed Controller
b) Electronic Speed Controller
c) Electric vehicle speed Controller
d) None of Above

**25.** By which Method we uploaded data to thingworx server from RPI

a) AMQP
b) LORA
c) REST
d) COAP

**26.** At what frequency Satellite transmits data to GPS?

a) GHZ
b) 1.4 GHZ
c) 2.4GHZ
d) none

**27.** On Which protocol RFID works

a) I2C
b) SPI
c) UART
d) none of above

**28.** Can we read analog Data with R-PI

a) NO
b) Yes
c) Maybe
d) none of above

**29.** GPS uses which hardware data transfer protocol to send data to controller

a) SPI
b) USART
c) UART
d) I2C

**30.** Which Actuator is used to control EV start and Stop

a) MOSFET's
b) Relay
c) Button
d) Key

**31.** The input voltage for raspberry pi model B is around _____

a) 5V
b) 10V
c) 20V
d) 40V

**32.** How many Usb ports does raspberry pi 3 model B+ contain?

a) 2
b) 4
c) 6
d) 3

**33.** The software pulse width modulation in raspberry pi 3 model B is available on _____ pins

a) GPIO3
b) GPIO18

c) GPIO12
d) All of Above

**34.** In which pin hardware pulse width modulation will not be available?

a) GPIO 12,13
b) GPIO 18,19
c) Both A and B
d) None of above

**35.** What is the standard form of SPI pin?

a) Serial Parallel Input
b) Serial Peripheral Interface
c) Serial Parallel Interface
d) None of the above

**36.** The I2C pin on the raspberry pi board has _____ connections

a) One
b) Two
c) Three
d) Four

**37.** _____ pins are the EEPROM pins on raspberry pi 3 model B

a) GPIO0
b) GPIO1
c) GPIO3
d) Both A and B

**38.** Which one uses a boot loader on a chip?

a) Arduino
b) Raspberry Pi

c) Both A and B
d) None of above

39. Which one of the following has integrated WIFI and Bluetooth?

a) Arduino
b) Raspberry Pi
c) Both A and B
d) None of above

40. What is the purpose of the ifconfig command?

a) Shows OS information
b) Shows past commands
c) Change the permission of the directory/file
d) Used to get the network information

41. SSH denotes

a) Secure Shell
b) Super user
c) Super host
d) None of mentioned

42. SUDO enables

a) Secure Shell
b) Super user
c) Super host
d) None of mentioned

43. _____ utility is usually used to check if communication can be made with another host.

a) Nmap
b) Ping

c) Ifconfig
d) None of the above

**44.** What will be the output of the following Python code?

```
1.>>>"abcd"[2:]
```

a) a
b) ab
c) cd
d) dc

**45.** What will be the output of the following Python statement?

```
1.>>>print('new' 'line')
```

a) Error
b) Output equivalent to print 'new\nline'
c) newline
d) new line

**46.** Which of the following is invalid?

a) _a = 1
b) __a = 1
c) __str__ = 1
d) none of the mentioned

**47.** Which of the following is not a keyword?

a) eval
b) assert
c) nonlocal
d) pass

**48.** What will be the output of the following Python code?

```
x = ['ab', 'cd']
for i in x:
   i.upper()
print(x)
```

a) [‘ab’, ‘cd’]
b) [‘AB’, ‘CD’]
c) [None, None]
d) none of the mentioned

**49.** What will be the output of the following Python code?

```
i = 1
while True:
   if i%3 == 0:
      break
   print(i)

   i + = 1
```

a) 1 2
b) 1 2 3
c) Error
d) none of the mentioned

**50.** What is the output when we execute list(“hello”)?

a) [‘h’, ‘e’, ‘l’, ‘l’, ‘o’]
b) [‘hello’]
c) [‘llo’]
d) [‘olleh’]

**51.** Suppose list1 is [2445,133,12454,123], what is max(list1)?

a) 2445

b) 133
c) 12454
d) 123

**52.** Suppose list1 is [2, 33, 222, 14, 25], What is list1[-1]?

a) Error
b) None
c) 25
d) 2

**53.** Which of the following statements create a dictionary?

a) d = {}
b) d = {"john":40, "peter":45}
c) d = {40:"john", 45:"peter"}
d) All of the mentioned

**54.** What will be the output of the following Python code snippet?

1 d = {"john":40, "peter":45}
a) "john", 40, 45, and "peter"
b) "john" and "peter"
c) 40 and 45
d) d = (40:"john", 45:"peter")

**55.** Which of the following is a Python tuple?

a) [1, 2, 3]
b) (1, 2, 3)
c) {1, 2, 3}
d) {}

**56.** What will be the output of the following Python code?

1.>>>t = (1, 2, 4, 3, 8, 9)

2.>>>[t[i] for i in range(0, len(t), 2)]

a) [2, 3, 9]
b) [1, 2, 4, 3, 8, 9]
c) [1, 4, 8]
d) (1, 4, 8)

**57.** What is the type of inf?

a) Boolean
b) Integer
c) Float
d) Complex

**58.** What does 3 ^ 4 evaluate to?

a) 81
b) 12
c) 0.75
d) 7

**59.** Given a function that does not return any value, What value is thrown by default when executed in shell.

a) Int
b) Bool
c) Void
d) None

**60.** What is the return type of function id?

a) int
b) Float
c) Bool
d) Dict

SET-A

Printed Pages: 09

University Roll No.....

**Mid Term Examination, Odd Semester 2020-21**

**B. Tech (All Branches), II Year, III Semester**

**Subject: English for Professional Purposes–I**

**Subject Code: BELH-0003**

**Time: 2 Hours** **Maximum Marks: 30**

**Instructions:**

A. It is mandatory to clearly mention the set number in your answer sheet.

B. Word limit for the questions in Section A is 60 while for the questions in Section B is 150.

**PART-A Communication Skills** **[TIME: 45 MINUTES]**

**Section- A** **3x2 = 6 Marks**

**Attempt all the questions.**

1. The underlined words in the following sentences are emphasized. What variation in the meaning will the emphasized words bring to these sentences?

A. I asked for a blue shirt.

B. I asked for a blue shirt.

C. I asked for a blue shirt.

2. See the cartoon and discuss what kind of barrier it will create for a Hindi speaker who knows that widows wear white dress in the Hindi speaking belt, but does not know anything about the Christian culture.

3. In the following sentence, underline the most likely thought group boundaries and write down the focus words in each thought group.

The first item on our agenda is to address parking.

**Section- B** **3x2 = 6 Marks**

**Attempt any two questions.**

4. Explain the process of communication with the help of a diagram.
5. Discuss the flow of communication and illustrate it with examples.

6. Read the essay given below and answer the following questions:
   A. Do you think this essay is well organized? Illustrate.
   B. Identify the transition markers and see whether they have been used correctly or not.

**Agriculture in India**

Agriculture is the oldest industry in India. Old books and scriptures mention it. Songs and prayers have been written for good harvest. In India, it has also been seen as the best occupation, better than any other. It gives employment to all sorts of people, trained or otherwise.

For centuries, it has been the backbone of the Indian economy. Until recently it used to employ over 80 percent of India's population. India produced enough for its own and other people. India has been an exporter of food grains. Actually, agriculture here was so good that the only people the British took out of India for their help are the farm labourers. They helped the British grow sugar cane and other commercial crops.

India grows a variety of crops. It produces a variety of cereals from rice to wheat, maize, millet, barley, corn, etc. Then there are different kinds of pulses. Of fruits and vegetables there are countless varieties. Commercial agriculture has also been popular in India. Besides tea, coffee and tobacco, it has always had sugarcane and spices which have brought foreigners and foreign exchange to India. Indian agriculture has always been eco-friendly, using only animals and their dung and vegetation to sustain productivity.

Everyone values the importance of agriculture in the economy of India. Universities and research institutions have been established to find ways to improve output. Union and state governments have ministries and departments of agriculture. Its place in the economy is unquestionable.

Yet in recent years, it has suffered. Talented youth no longer comes to agriculture. Besides, governments have done little to bring modern methods of farming in India. Old methods of irrigation and flood control have disappeared. As a result, Indian agriculture has faced problems like suicides by debt-burdened farmers, and years of droughts and floods, resulting in crop failures.

We must bring talent back into agriculture. We must not limit the size and holding of plots. Farmers should be allowed to own and expand like industries. Unreasonable legislations like land ceiling act must go, farm labour must be better paid and better treated. Farm insurance must come to farmers. Crops die for want of water, even by the side of rivers. Solar power can help irrigate farms. Rivers must be allowed to flow freely. Farm machinery continues to be unaffordable and unsuitable in India.

We must make farm machinery answer our needs. They cost less and work better on both small and big plots of land. Agriculture marketing must be more business-like. Famers do not need subsidy, they need freedom to compete and charge competitive prices.

With these changes, as the famous agriculture scientist, Dr M S Swaminathan says, Indians can feed the world. It will attract talented and enterprising people and will reward them with respectable returns. Earth is seen as mother in India. And mother can never fail. All that is required is giving agriculture its due place. Rewards can be incalculable.

## PART B- VERBAL APTITUDE [TIME: 75 MINUTES]

**Attempt all questions. Each question carries 0.25 mark.**
**Choose the correct option to fill in the blanks. (Q 1-46)**

1. It was all fake news that he was made ..... by the custom officer.
A. to show his passport
B. show his passport
C. shown his passport
D. showed his passport
E. showing his passport

2. While crossing the bridge, I saw him ..... from it.
A. jumping
B. to jump
C. jumped
D. jump
E. to jumping

3. I prefer ..... at home rather than ..... to that city.
A. staying, to go
B. to stay, go
C. stayed, to go
D. stay, going
E. to Stay, going

4. An exemplary penalty should be ..... for ..... undue adjournments and initiating frivolous litigation.
A. impose, seeking
B. imposed, seek
C. imposed, sought
D. imposing, seeking
E. imposed, seeking

5. She heard the doorbell ..... to answer it.
A. rang and gone
B. ring and went
C. ringing and went
D. rang and going
E. rang and went

6. She is quite nervous. I think, she ..... the answer.
A. is not knowing
B. does not know
C. do not know
D. did not knew
E. was not knowing

7. The bus ___ left the station before we ___.
A. has, reached
B. has been, reached
C. had, reach
D. had, reached
E. has, reach

8. I ___ while my brother was making yakhni for us.
A. slept
B. am sleeping
C. was sleeping
D. has been sleeping
E. sleep

9. I have applied medicine to the wound. It ___ be fine by tomorrow.
A. can
B could
C. need
D. dare
E. should

10. I have already spent the ___ money I had in my account.
A. few
B. some
C. little
D. all
E. all

11. You have to face __ challenges before you get a job.
A. any
B. little
C. neither
D. either
E. many

12. Mohit has not brought ___ food for the picnic.
A. some
B. little
C. few
D. any
E. all

13. Though my friend has ___ knowledge about the subject she passed the exam.
A. few
B. little
C. enough
D. a lot of
E. lots of

14. ___ mango is ripe.
A. Neither
B. All
C. Both
D. A few
E. The few

15. __ poverty still persists in our country even decades after independence.
A. A great number of
B. Many
C. Much
D. Lots of
E. Few

16. Bhavya said, "___ a beautiful garden it is!"
A. This
B. That
C. Very
D. Really
E. What

17. He doesn't have ___ friends.
A. few
B. many
C. some
D. all
E. both

18. Mrs. Bose is asking if she might have __ moment with Mrs. Sharma in private.
A. a
B. some
C. a few
D. a little
E. any

19. There is .....milk left in the container; therefore, only if the child is very hungry give it to him.
A. much
B. little
C. few
D. any
E. a lot

20. ___ of what he stated was intelligible to us.
A. Many
B. A few
C. Much
D. Neither
E. Either

21. We have bought a new machine for our mother to help her with the household chores, the machine is _____.
A. cartable
B. portable
C. potable
D. relatable
E. retable

22. We need a _____ to decode these cryptic maps.
A. calligrapher
B. cartographer
C. choreographer
D. curator
E. cursor

23. I am a/an _____ who looks at the dark side of everything.
A. naturalist
B. chauffeur
C. numismatist
D. optimist
E. pessimist

24. They used him as a _____ to accuse him of the whole conspiracy.
A. scapedog

B. scapecat
C. scapegoat
D. scapecow
E. scapesheep

25. Everyone in the meeting praised Ram for his presentation. Suddenly Kavita from behind stood up to oppose because she is _____
A. a protagonist
B. a sidekick
C. an antagonist
D. a comrade
E. a sadist

26. Jacob's cousin will surely disclose the secret of Jacob's somnambulism in college as he loves to see Jacob being humiliated. Jacob's cousin is _____
A. a shrewd
B. a sadist
C. a truant
D. a masochist
E. a tyrant

27. The doctor has put him for a blood purification process as his kidneys malfunction. He is under_____.
A. dialysis
B. ventilation
C. integration
D. catharsis
E. cardiopulmonary

28. Sameera prefers to travel on foot for short distances. She is a _____.
A. pedlar
B. pedestrian
C. jogger
D. stickler
E. runner

29. Mr. Gupta is an epitome of population explosion; he has four wives and 12 children from them. He is a _____.
A.bigamist
B.polygamist
C. monogamist
D.both a and c
E. digamist

30. Raina asked me to help him for ___ days.
A. a few
B. few
C. a little
D. little
E. a great deal of

31. ___ window is repaired.
A. Both
B. Neither
C. All
D. What
E. Any

32. She said she was not editing ____ paper for the presentation.
A. some
B. all
C. such
D. half
E. any

33. Some of the milk_____ still there.
A. is
B. are
C. has
D. have
E. is to

34. Measles _____ a common disease.
A. are
B. is
C. were
D. has
E. have

35. Do you _____ these activities?
A. liking
B. likes
C. like
D. have like
E. have liked

36. The courier _____ arrived.
A. have
B. are
C. were
D. has

E. have to be

37. Either of the two candidates _____ eligible.
A. is
B. have
C. are
D. has been
E. have been

38. _____ the plan been finalized?
A. Have
B. Has
C. Is
D. Are
E. was

39. They _____ in the tournament to represent their university.
A. participates
B. was participating
C. participating
D. have participate
E. are participating

40. The committee _____ various solutions.
A. have
B. was
C. are
D. is
E. were

41. The manager and the assistant _____ together.
A. enters
B. has entered
C. have entered
D. had enter
E. has been entering

42. One of the students _____ to clear this examination.
A. failing
B. fails
C. fail
D. has failed
E. have failed

43. Neither she nor her brother _____ involved in cheating.
A. was
B. are
C. has
D. were
E. involved

44. This set of questions _____ very easy.
A. are
B. is
C. was
D. have
E. has

45. Time and tide _____ for none.
A. waiting
B. wait
C. waits
D. waited
E. to wait

46. Your trousers _____ there on the floor.
A. is
B. has
C. are
D. was
E. have

**Choose the grammatically correct sentence. (Q. 47-56)**

47. A. We hereby send you the signing contract.
B. We hereby send you to signing contract.
C. We hereby send you the signed contract.
D. We hereby sending you the signing contract.
E. We hereby sending you the signed contract.

48. A. Many parents avoided sending their kids to school in Covid- 19.
B. Many parents avoided to send their kids to school in Covid- 19.
C. Many parents avoided send their kids to school in Covid- 19.
D. Many parents avoided to sending their kids to school in Covid- 19.
E. Many parents avoided sent their kids to school in Covid- 19.

49. A. Over a period of time, I have found that our judiciary has taken note of these suggestions and started their implementation in phases.
B. Over a period of time, I have found that our judiciary has taken note of these suggestions and have started their implementation in phases.
C. Over a period of time, I have found that our judiciary has taken note of these suggestions and has started their implementation in phases.
D. Over a period of time, I have found that our judiciary has taken note of these suggestions and has been starting their implementation in phases.
E. Over a period of time, I have found that our judiciary has taken note of these suggestions and have been starting their implementation in phases.

50. A. The submarine dived to a depth of about 30 meters and started to cruise.
B. The submarine dived to a depth of about 30 meters and started cruise.
C. The submarine dived to a depth of about 30 meters and started cruised.
D. The submarine dived to a depth of about 30 meters and started to cruising.
E. The submarine dived to a depth of about 30 meters and started cruising.

51. A. What could you like to have –tea, coffee or coke? I must rather have a glass of warm water than anything else.
B. What would you like to have –tea, coffee or coke? I would rather have a glass of warm water than anything else.
C. What shall you like to have –tea, coffee or coke? I might rather have a glass of warm water than anything else.
D. What should you like to have –tea, coffee or coke? I could rather have a glass of warm water than anything else.
E. What might you like to have –tea, coffee or coke? I should rather have a glass of warm water than anything else.

52. A. No sooner had my mother been heard the news than she rushed out of the house.
B. No sooner than my mother had heard the news she rushed out of the house.
C. No sooner had my mother heard the news than she rushed out of the house.
D. My mother rushed out of the house no sooner than she heard the news.
E. My mother heard the news no sooner than she rushed out of the house.

53. A. I don't know the number of the files which is to be written on the envelop.
B. I don't know a few number of the files which is to be written on the envelop.
C. I don't know a number of the files which are to be written on the envelop.
D. I don't know a number of the files which are to be written in the envelop.
E. I didn't know a numbers of the files which are to be written on the envelop.

54. A. Do the colour of the rainbow distinctly visible from here?
B. Does the colours of the rainbow distinctly visible from here?
C. Do the colours of the rainbow distinctly visible from here?
D. Are the colours of the rainbow distinctly visible from here?
E. Are the colour of the rainbows distinctly visible from here?

55. A. I want the first three books in that row.
B. I want first the three books in that row.
C. I want the three first books in that row.
D. I want first three the books in that row.
E. I want the first books three in that row.

56. A. One-third student came to the event.
B. One-third of the student came to the event.
C. One-third of the students came to the event.
D. One-third the student came to the event.
E. The one-third student came to the event.

57. If you want a cup of coffee from your friend, you would say
A. "Get me a cup of coffee, will you?"
B. "Get me a cup of coffee, do you?"
C. "Get me a cup of coffee, are you?"
D. "Get me a cup of coffee, will I?"
E. "Get me a cup of coffee, do I?"

58. "Govt." is an example of
A. Blending
B. Compounding
C. Clipping
D. Derivation
E. Inflexion

59. Which among the following sentences carries an intransitive verb?
A. A flock of birds is flying in the sky.
B. He is flying a kite in the sky.
C. Adarsh had successfully flown the aircraft.
D. They flew the flag of India at military head quarter.
E. See a fly on the wall.

60. Which among the following sentences carries a ditransitive verb?
A. The vice- chancellor delivered the speech in the hall.
B. The leader of the opposition criticized the government at the press conference.
C. The general manager asked him difficult questions during the interview.
D. She sleeps till seven in the morning.
E. Rishab will become pilot next year.

61. In which sentence the word 'blind' is used as a verb?
A. He is blind but he can read and speak eight different languages.
B. Strong rays of sunlight may blind you. Come inside the room.
C. There is enough light inside, could you please turn off these binds please?
D. Sushobhit groped blindly for his glasses on the table.
E. He declared that the verdict was the result of blind prejudice.

62. Identify the part of speech of 'that' in the sentence written below.
Akanksha plays violin really well, but she is not that good at piano.
A. Adjective
B. Adverb
C. Preposition
D. Pronoun
E. Conjunction

63. What was Krishna doing in the park today? The grammatically correct answer of this question would be
A. Exercise was Krishna doing in the park today
B. Krishna doing exercise in the park today
C. Krishna did exercise in the park today
D. Krishna was doing exercise in the park today
E. Exercise was doing Krishna in the park today

64. If we use a present participle in the sentence: "Do you have a bag which folds?", the sentence will be
A. Do you have a folding bag?
B. Do you have a bag which can be folded?
C. Do you have a bag which is folded?
D. Do you have a folded bag?
E. Do you have a fold bag?

65. Choose the sentence with correct punctuation marks.
A. "Don't you dare talk to me like that", said my teacher.
B. "Dont you dare talk to me like that", said my teacher.
C. Don't you dare talk to me like that, said my teacher.
D. Dont you dare talk to me like that said my teacher.
E. Don't you dare talk to me like that, said my teacher.

**Given below are sentences with underlined phrases/clauses. Choose the best word to substitute the phrase/clause. (Q 66-68)**

66. I don't mind talking to him about the project but he is <u>a person who thinks only of himself.</u>
A. an eccentric
B. a boaster
C. an egocentric
D. a proud
E. an introvert

67. My sister is wildly <u>unreasonable, illogical and ridiculous.</u>
A. pantomime
B. absurd

C. abject
D. arid
E. adept

68. Salma's father is an author and has won an award for writing a book that talks about his own life story.
A. a bibliography
B. a biography
C. an autobiography
D. a tomography
E. a geography

**Select the correct subject-verb combination in the following sentences. (Q 69-72)**

69. Naresh along with his seniors is preparing to participate in an international conference.
A. 'his seniors' and 'preparing'
B. 'Naresh' and 'is preparing'
C. 'Naresh' and 'preparing'
D. 'Naresh' and 'his seniors'
E. 'Naresh' and 'to prepare'

70. Three fourths of the work is done by Naina herself.
A. 'Three fourths' and 'is'
B. 'Three fourths' and 'done'
C. 'Three fourths' and 'herself'
D. 'Three fourths of the work' and 'is done'
E. 'Three fourths of works' and 'is done'

71. There is a box of mangoes in the yellow car near the building.
A. 'There' and 'is'
B. 'a box' and 'mangoes'
C. 'a box of mangoes' and 'is'
D. 'There' and 'car'
E. 'There is' and 'near the building'

72. None but the class representative of the class has submitted the assignment.
A. "None" and "have submitted"
B. "the class representative" and "has submitted"
C. "the class" and "have"
D. "class representative" and "assignment"
E. "None" and "class representative is"

SET-B

Printed Pages : 09

University Roll No.....

Mid Term Examination, Odd Semester 2020-21

B. Tech (All Branches), II Year, III Semester

Subject: English for Professional Purposes–I

Subject Code: BELH-0003

Time: 2 Hours

Maximum Marks: 30

Instructions:

A. It is mandatory to clearly mention the set number in your answer sheet.

B. Word limit for the questions in Section A is 60 while for the questions in Section B is 150.

## PART-A Communication Skills [TIME: 45 MINUTES]

### Section- A

3x2 = 6 Marks

**Attempt all the questions.**

1. The underlined words in the following sentences are emphasized. What variation in the meaning will the emphasized words bring to these sentences?

   A. I asked for a **blue** shirt.

   B. **I** asked for a blue shirt.

   C. I asked for a blue **shirt.**

2. In the following sentence, underline the most likely thought group boundaries and write down the focus words in each thought group.

   The first item on our agenda is to address parking.

3. See the cartoon and discuss what kind of barrier it will create for a Hindi speaker who knows that widows wear white dress in the Hindi speaking belt, but does not know anything about the Christian culture.

## Section- B 3x2 = 6 Marks

**Attempt any two questions.**

4. Discuss the flow of communication and illustrate it with examples.
5. Explain the process of communication with the help of a diagram.
6. Read the essay given below and answer the following questions:
   A. Do you think this essay is well organized? Illustrate.
   B. Identify the transition markers and see whether they have been used correctly or not.

### Agriculture in India

Agriculture is the oldest industry in India. Old books and scriptures mention it. Songs and prayers have been written for good harvest. In India, it has also been seen as the best occupation, better than any other. It gives employment to all sorts of people, trained or otherwise.

For centuries, it has been the backbone of the Indian economy. Until recently it used to employ over 80 percent of India's population. India produced enough for its own and other people. India has been an exporter of food grains. Actually, agriculture here was so good that the only people the British took out of India for their help are the farm labourers. They helped the British grow sugar cane and other commercial crops.

India grows a variety of crops. It produces a variety of cereals from rice to wheat, maize, millet, barley, corn, etc. Then there are different kinds of pulses. Of fruits and vegetables there are countless varieties. Commercial agriculture has also been popular in India. Besides tea, coffee and tobacco, it has always had sugarcane and spices which have brought foreigners and foreign exchange to India. Indian agriculture has always been eco-friendly, using only animals and their dung and vegetation to sustain productivity.

Everyone values the importance of agriculture in the economy of India. Universities and research institutions have been established to find ways to improve output. Union and state governments have ministries and departments of agriculture. Its place in the economy is unquestionable.

Yet in recent years, it has suffered. Talented youth no longer comes to agriculture. Besides, governments have done little to bring modern methods of farming in India. Old methods of irrigation and flood control have disappeared. As a result, Indian agriculture has faced problems like suicides by debt-burdened farmers, and years of droughts and floods, resulting in crop failures.

We must bring talent back into agriculture. We must not limit the size and holding of plots. Farmers should be allowed to own and expand like industries. Unreasonable legislations like land ceiling act must go, farm labour must be better paid and better treated. Farm insurance must come to farmers. Crops die for want of water, even by the side of rivers. Solar power can help irrigate farms. Rivers must be allowed to flow freely. Farm machinery continues to be unaffordable and unsuitable in India.

We must make farm machinery answer our needs. They cost less and work better on both small and big plots of land. Agriculture marketing must be more business-like. Famers do not need subsidy, they need freedom to compete and charge competitive prices.

With these changes, as the famous agriculture scientist, Dr M S Swaminathan says, Indians can feed the world. It will attract talented and enterprising people and will reward them with respectable returns. Earth is seen as mother in India. And mother can never fail. All that is required is giving agriculture its due place. Rewards can be incalculable.

## PART B- VERBAL APTITUDE [TIME: 75 MINUTES]

**Attempt all questions. Each question carries 0.25 mark.**
**Choose the correct option to fill in the blanks. (Q 1-46)**

1. I prefer ..... at home rather than ..... to that city.
A. staying, to go
B. to stay, go
C. stayed, to go
D. stay, going
E. to stay, going

2. She heard the doorbell ..... to answer it.
A. rang and gone
B. ring and went
C. ringing and went
D. rang and going
E. rang and went

3. The bus .....left the station before we .....
A. has, reached
B. has been, reached
C. had, reach
D. had, reached
E. has, reach

4. I have applied medicine to the wound. It ..... be fine by tomorrow.
A. can
B. could
C. need
D. dare
E. should

5. While crossing the bridge, I saw him ..... from it.
A. jumping
B. to jump
C. jumped
D. jump
E. to jumping

6. I have already spent the ..... money I had in my account.
A. few
B. some
C. little
D. all
E. all

7. An exemplary penalty should be ..... for ..... undue adjournments and initiating frivolous litigation.
A. impose, seeking
B. imposed, seek
C. imposed, sought
D. imposing, seeking
E. imposed, seeking

8. She is quite nervous. I think, she ..... the answer.
A. is not knowing
B. does not know
C. do not know
D. did not knew
E. was not knowing

9. You have to face ..... challenges before you get a job.
A. any
B. little
C. neither
D. either
E. many

10. I ..... while my brother was making Yakhni for us.
A. slept
B. am sleeping
C. was sleeping
D. has been sleeping
E. sleep

11. Mohit has not brought ..... food for the picnic.
A. some
B. little
C. few
D. any
E. all

12. Mr. Gupta is an epitome of population explosion; he has four wives and 12 children from them. He is a .....
A. bigamist
B. polygamist
C. monogamist
D. both a and c
E. digamist

13. There is .....milk left in the container; therefore, only if the child is very hungry give it to him.
A. much
B. little
C. few
D. any
E. a lot

14. We have bought a new machine for our mother to help her with the household chores, the machine is .....
A. cartable
B. portable
C. potable
D. relatable
E. retable

15. ..... poverty still persists in our country even decades after independence.
A. A great number of
B. Many
C. Much
D. Lots of
E. Few

16. He doesn't have ..... friends.
A. few
B. many
C. some
D. all
E. both

17. Though my friend has ..... knowledge about the subject she passed the exam.
A. few
B. little
C. enough
D. a lot of
E. lots of

18. Mrs. Bose is asking if she might have ..... moment with Mrs. Sharma in private.
A. a
B. some
C. a few
D. a little
E. any

19. .....mango is ripe.
A. Neither
B. All
C. Both
D. A few
E. The few

20. .....of what he stated was intelligible to us.
A. Many
B. A few
C. Much
D. Neither
E. Either

21. We need a .....to decode these cryptic maps.
A. calligrapher
B. cartographer
C. choreographer
D. curator
E. cursor

22. Bhavya said, ".....a beautiful garden it is!"
A. This
B. That
C. Very
D. Really
E. What

23. I am a/an ..... who looks at the dark side of everything.
A. naturalist
B. chauffeur
C. numismatist
D. optimist
E. pessimist

24. Everyone in the meeting praised Ram for his presentation. Suddenly Kavita from behind stood up to oppose because she is .....
A. a protagonist
B. a sidekick
C. an antagonist
D. a comrade
E. a sadist

25. Raina asked me to help him for .....days.
A. a few
B. few
C. a little
D. little
E. a great deal of

26. Jacob's cousin will surely disclose the secret of Jacob's somnambulism in college as he loves to see Jacob being humiliated. Jacob's cousin is .....
A. a shrewd
B. a sadist
C. a truant
D. a masochist
E. a tyrant

27. Some of the milk..... still there.
A. is
B. are
C. has
D. have
E. is to

28. Sameera prefers to travel on foot for short distances. She is a .....
A. pedlar
B. pedestrian
C. jogger
D. stickler
E. runner

29. Either of the two candidates .....eligible.
A. is
B. have
C. are
D. has been
E. have been

30. .....window is repaired.
A. Both
B. Neither
C. All
D. What
E. Any

31. They used him as a .....to accuse him of the whole conspiracy.
A. scapedog
B. scapecat
C. scapegoat
D. scapecow
E. scapesheep

32. She said she was not editing ..... paper for the presentation.
A. some
B. all
C. such
D. half
E. any

33. The doctor has put him for a blood purification process as his kidneys malfunction. He is under.....
A. dialysis
B. ventilation
C. integration
D. catharsis
E. cardiopulmonary

34. Measles ..... a common disease.
A. are
B. is
C. were
D. has
E. have

35. Do you ..... these activities?
A. liking
B. likes
C. like
D. have like
E. have liked

36. The courier .....arrived.
A. have
B. are
C. were
D. has
E. have to be

37. Your trousers ..... there on the floor.
A. is
B. has
C. are
D. was
E. have

38. .....the plan been finalized?
A. Have
B. Has
C. Is
D. Are
E. Was

39. One of the students ..... to clear this examination.
A. failing
B. fails
C. fail
D. has failed
E. have failed

40. They ..... in the tournament to represent their university.
A. participates
B. was participating
C. participating
D. have participate
E. are participating

41. The manager and the assistant ..... together.
A. enters
B. has entered
C. have entered
D. had enter
E. has been entering

42. This set of questions ..... very easy.
A. are
B. is
C. was
D. have
E. has

43. Neither she nor her brother ..... involved in cheating.
A. was
B. are
C. has
D. were
E. involved

44. It was all fake news that he was made .....by the custom officer.
A. to show his passport
B. show his passport
C. shown his passport
D. showed his passport
E. showing his passport

45. Time and tide ..... for none.
A. waiting
B. wait
C. waits
D. waited
E. to wait

46. The committee ..... various solutions.
A. have
B. was
C. are
D. is
E. were

**Choose the grammatically correct sentence. (Q 47-56)**

47. A. Many parents avoided sending their kids to school in Covid- 19.
B. Many parents avoided to send their kids to school in Covid- 19.
C. Many parents avoided send their kids to school in Covid- 19.
D. Many parents avoided to sending their kids to school in Covid- 19.
E. Many parents avoided sent their kids to school in Covid- 19.

48. A. The submarine dived to a depth of about 30 meters and started to cruise.
B. The submarine dived to a depth of about 30 meters and started cruise.
C. The submarine dived to a depth of about 30 meters and started cruised.
D. The submarine dived to a depth of about 30 meters and started to cruising.
E. The submarine dived to a depth of about 30 meters and started cruising.

49. A. Do the colour of the rainbow distinctly visible from here?
B. Does the colours of the rainbow distinctly visible from here?
C. Do the colours of the rainbow distinctly visible from here?
D. Are the colours of the rainbow distinctly visible from here?
E. Are the colour of the rainbows distinctly visible from here?

50. A. We hereby send you the signing contract.
B. We hereby send you to signing contract.
C. We hereby send you the signed contract.
D. We hereby sending you the signing contract.
E. We hereby sending you the signed contract.

51. A. What could you like to have –tea, coffee or coke? I must rather have a glass of warm water than anything else.
B. What would you like to have –tea, coffee or coke? I would rather have a glass of warm water than anything else.
C. What shall you like to have –tea, coffee or coke? I might rather have a glass of warm water than anything else.
D. What should you like to have –tea, coffee or coke? I could rather have a glass of warm water than anything else.
E. What might you like to have –tea, coffee or coke? I should rather have a glass of warm water than anything else.

52. A. No sooner had my mother been heard the news than she rushed out of the house.
B. No sooner than my mother had heard the news she rushed out of the house.
C. No sooner had my mother heard the news than she rushed out of the house.
D. My mother rushed out of the house no sooner than she heard the news.
E. My mother heard the news no sooner than she rushed out of the house.

53. A. I don't know the number of the files which is to be written on the envelop.
B. I don't know a few number of the files which is to be written on the envelop.
C. I don't know a number of the files which are to be written on the envelop.
D. I don't know a number of the files which are to be written in the envelop.
E. I didn't know a numbers of the files which are to be written on the envelop.

54. A. Over a period of time, I have found that our judiciary has taken note of these suggestions and started their implementation in phases.
B. Over a period of time, I have found that our judiciary has taken note of these suggestions and have started their implementation in phases.
C. Over a period of time, I have found that our judiciary has taken note of these suggestions and has started their implementation in phases.
D. Over a period of time, I have found that our judiciary has taken note of these suggestions and has been starting their implementation in phases.
E. Over a period of time, I have found that our judiciary has taken note of these suggestions and have been starting their implementation in phases.

55. A. One-third student came to the event.
B. One-third of the student came to the event.
C. One-third of the students came to the event.
D. One-third the student came to the event.
E. The one-third student came to the event.

56. If you want a cup of coffee from your friend, you would say
A. "Get me a cup of coffee, will you?"
B. "Get me a cup of coffee, do you?"
C. "Get me a cup of coffee, are you?"
D. "Get me a cup of coffee, will I?"
E. "Get me a cup of coffee, do I?"

57. "Govt." is an example of
A. Blending
B. Compounding
C. Clipping
D. Derivation
E. Inflexion

58. Which among the following sentences carries an intransitive verb?
A. A flock of birds is flying in the sky.
B. He is flying a kite in the sky.
C. Adarsh had successfully flown the aircraft.
D. They flew the flag of India at military head quarter.
E. See a fly on the wall.

59. If we use a present participle in the sentence: "Do you have a bag which folds?", the sentence will be
A. Do you have a folding bag?
B. Do you have a bag which can be folded?
C. Do you have a bag which is folded?
D. Do you have a folded bag?
E. Do you have a fold bag?

60. Which among the following sentences carries a ditransitive verb?
A. The vice- chancellor delivered the speech in the hall.
B. The leader of the opposition criticized the government at the press conference.
C. The general manager asked him difficult questions during the interview.
D. She sleeps till seven in the morning.
E. Rishab will become pilot next year.

61. In which sentence the word 'blind' is used as a verb?
A. He is blind but he can read and speak eight different languages.
B. Strong rays of sunlight may blind you. Come inside the room.
C. There is enough light inside, could you please turn off these binds please?
D. Sushobhit groped blindly for his glasses on the table.
E. He declared that the verdict was the result of blind prejudice.

62. A. I want the first three books in that row.
B. I want first the three books in that row.
C. I want the three first books in that row.
D. I want first three the books in that row.
E. I want the first books three in that row.

63. Choose the sentence with correct punctuation marks.
A. "Don't you dare talk to me like that", said my teacher.
B. "Dont you dare talk to me like that", said my teacher.
C. Don't you dare talk to me like that, said my teacher.
D. Dont you dare talk to me like that said my teacher.
E. Don't you dare talk to me like that, said my teacher.

64. Identify the part of speech of 'that' in the sentence written below.
What was Krishna doing in the park today?
The grammatically correct answer of this question would be
A. Exercise was Krishna doing in the park today
B. Krishna doing exercise in the park today
C. Krishna did exercise in the park today
D. Krishna was doing exercise in the park today
E. Exercise was doing Krishna in the park today

65. Akanksha plays violin really well, but she is not that good at piano.
A. Adjective

B. Adverb
C. Preposition
D. Pronoun
E. Conjunction

**Given below are sentences with underlined phrases/clauses. Choose the best word to substitute the phrase/clause. (Q 66-68)**

66. I don't mind talking to him about the project but he is a person who thinks only of himself.
A. an eccentric
B. a boaster
C. an egocentric
D. a proud
E. an introvert

67. Salma's father is an author and has won an award for writing a book that talks about his own life story.
A. a bibliography
B. a biography
C. an autobiography
D. a tomography
E. a geography

68. My sister is wildly unreasonable, illogical and ridiculous.
A. pantomime
B. absurd
C. abject
D. arid
E. adept

**Select the correct subject-verb combination in the following sentences. (Q 69-72)**

69. Three fourths of the work is done by Naina herself.
A. 'Three fourths' and 'is'
B. 'Three fourths' and 'done'
C. 'Three fourths' and 'herself'
D. 'Three fourths of the work' and 'is done'
E. 'Three fourths of works' and 'is done'

70. Naresh along with his seniors is preparing to participate in an international conference.
A. 'his seniors' and 'preparing'
B. 'Naresh' and 'is preparing'
C. 'Naresh' and 'preparing'
D. 'Naresh' and 'his seniors'
E. 'Naresh' and 'to prepare'

71. None but the class representative of the class has submitted the assignment.
A. "None" and "have submitted"
B. "the class representative" and "has submitted"
C. "the class" and "have"
D. "class representative" and "assignment"
E. "None" and "class representative is"

72. There is a box of mangoes in the yellow car near the building.
A. 'There' and 'is'
B. 'a box' and 'mangoes'
C. 'a box of mangoes' and 'is'
D. 'There' and 'car'
E. 'There is' and 'near the building'